

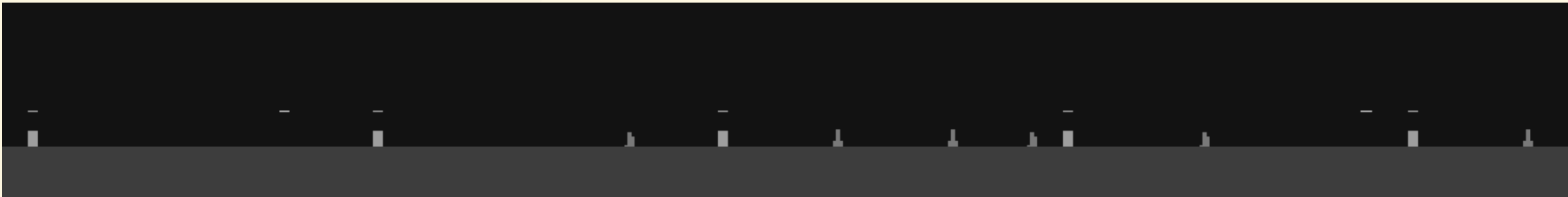
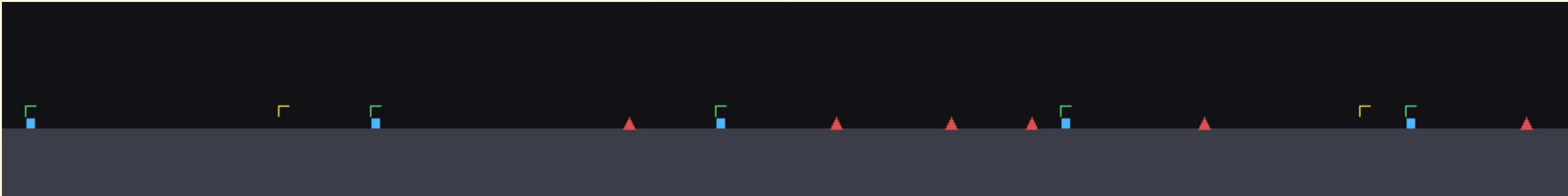
Geometry Dash Vision RL Agent

CS231N Milestone 3 • Preliminary Results

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Environment Validation

Goal. Train a vision-only agent that maps one rendered game frame to jump/no-jump.



Preliminary Evaluation Setup

Curriculum. Five simple levels: flat, one spike, two spikes, late spike, mixed spikes.

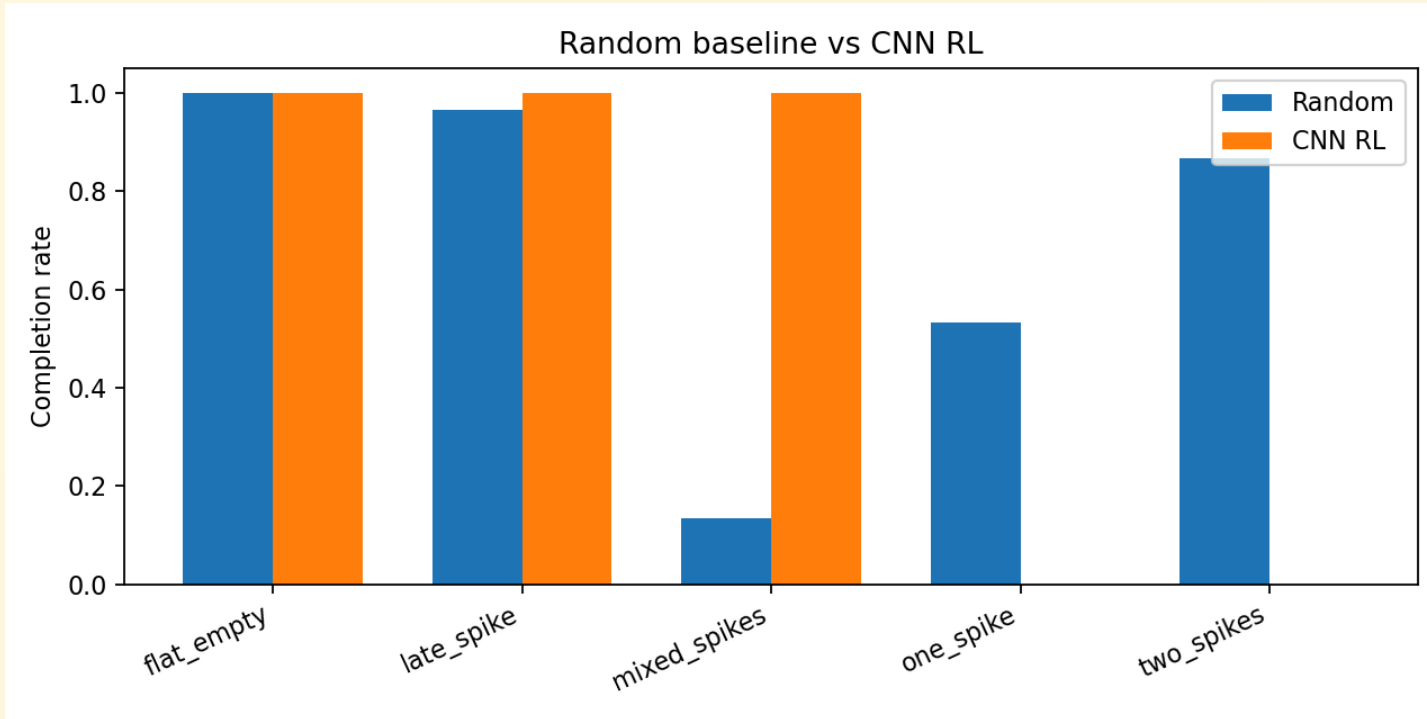
Observation. Single grayscale image frame, shaped `(1, 96, 96)`.

Policy. Tiny CNN trained with REINFORCE.

Baseline. Random binary policy, 30 episodes per level.

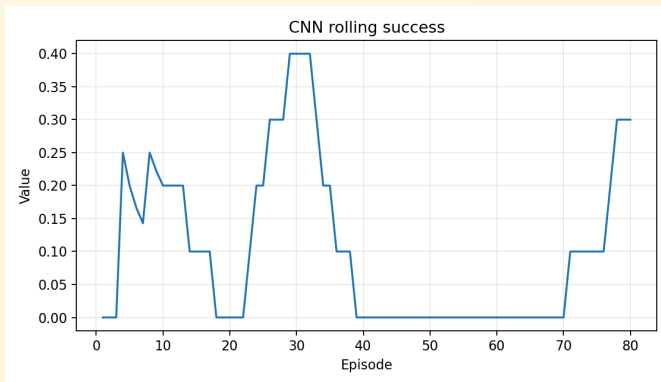
Metrics. Completion rate, average progress, average reward.

Results: Random vs CNN RL

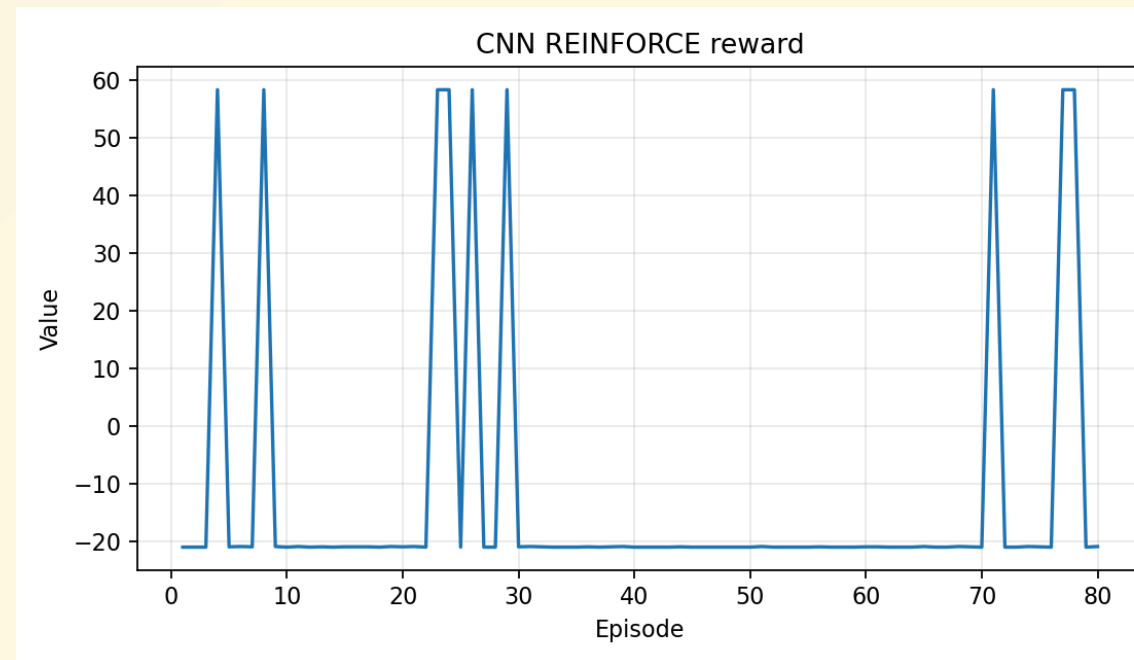


Finding. Random is surprisingly strong on very easy levels because repeated jump attempts often clear simple spike layouts.

Training Dynamics



Takeaway. REINFORCE sometimes succeeds but remains unstable; sparse timing rewards are the bottleneck.



Analysis and Limitations

Working so far.

- Live visual observations, deterministic rollouts, and reproducible evaluation are working.
- Physics corrections are covered by targeted regression tests.

Current limits.

- CNN + REINFORCE is too noisy for stable jump timing.
- Random remains competitive because the starter levels are very forgiving.

Next Steps

1. Add short frame history or optical-flow-like velocity cues.
2. Replace vanilla REINFORCE with PPO or actor-critic.
3. Tune curriculum so random is weak but solvable behavior remains clear.
4. Add rollout visualizations for learned policies.
5. Report mean $\pm$ std over multiple seeds.

Milestone takeaway. The environment and evaluation harness are working; the learning algorithm is the current bottleneck.